Neural Graphical Models for Healthcare Data: Representation, Learning, and Probabilistic Inference

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***Abstract*—This project aims to implement Neural Graphical Models(NGMs) that combine graphical dependency structure with nonlinear learning capability of neural networks. The project mainly focuses on three pillars: Representation, Learning and Inference. First, the relationship between variables will be established using a graphical structure and neural network. The model will then learn these relationships from the data. The learned model will predict unknown variables from known information. Finally, the model will be validated on real world and synthetic data.**

***Index Terms*—Probabilistic Graphical Model(PGM), Neural Network, Neural Graphical Models(NGMs).**

I. Introduction

Conventional Probabilistic Graphical Models(PGMs) such as Bayesian Networks are very useful to interpret that how different variables are related and also for making predictions using probabilities. However, traditional PGMs can face some limitations in modeling complex nonlinear relationship and diversified data. Machine learning along with PGM can solve such limitations as the model learn complex patterns from real world data while keeping relationship between variables understandable. This project aims to implement a Neural Graphical Model(NGMs) that combines the structured dependency representation of graphical models with the nonlinear learning capability of artificial neural networks.

1. Representation: representing graphical dependencies through neural network so the model uses learned weights to show how strongly on variable is related to another
2. Learning: Neural network help the model to understand complex and nonlinear relationship while maintain consistency with the specified graph structure.
3. Inference: Trained model can predict unknown variables from the variables that are already know. Means the trained Neural Graphical Model estimates the most likely values of unknown variables.

II. Project Summary

Probabilistic Graphical Models(PGMs) uses graphs to show the relationship between different variables. They use probability to represent uncertainty and predict unknown variables based on the known information. But the conventional graphical models may not efficiently capture complex nonlinear relationships. In this project we implement Neural Graphical Model(NGMs) based on the base paper "Neural Graphical Model" by Harsh Shrivastava and Urszula Chajewska[1]. The project mainly emphasis on the three fundamental pillars of NGM: representation learning and inference. During the representation stage a given graphical structure will be represented by a neural network where paths and learned weights capture dependencies between the variables. During the learning stage the neural network will learn nonlinear relationships from data while the soft graph constraint helps the network to follow the relationships given in the original path. In the last stage Inference stage selected variables will be observed while unknown variables will be predicted using the trained NGM through gradient-based optimization. In the validation phase first the model will be validated using synthetically generated graphical model data with known dependency structure. Then the model will be evaluated on a real world dataset containing both numeric and categorical variables. Now at last the NGM will be compared with the traditional approaches like Bayesian networks and Logistic regression. The aim of the project is to demonstrate that how neural networks can improve the flexibility their structured and interpretable nature.

III. Key Objectives

Here are the key objectives, we are claiming to achieve with study in this domain area with relevant research papers, datasets and codes.

- Implementing a Neural Graphical Model (NGM) for representing complex nonlinear dependencies among healthcare-related variables.
- Learning the parameters and dependency relationships of the Neural Graphical Model from the dataset while preserving the specified graphical structure.
- Estimating the likelihood of lung cancer and other clinical variables through probabilistic inference given patient information.
- Comparing the Neural Graphical Model against appropriate baseline methods using relevant Key Performance Indicators (KPIs).
- Analyzing dependency graphs and the Neural Graphical Model (NGM)-based approach in the context of state-of-the-art (SOTA) research from Microsoft Research, and identifying its different aspects and applications in the healthcare domain.

IV. Expected Outcomes

The expected outcome of this project are:
1. Neural Graphical Model implementation Develop NGM that contains Graphical Models (represent relationship between variables) and Neural network (to learn complex nonlinear relationships from data).
2. Representation of Dependencies We can show the relationships between the variables in graphical model using neural network connections.
3. Learning of Nonlinear relationships In this stage will train the model to learn the nonlinear relationships from the dataset while maintaining consistency with the given graph structure.
4. Inference on unknown variables In inference method will use the trained NGM to predict unknown variables from the available information
5. Experimental evaluation To study model's representation learning and inference performance will test on a synthetic dataset with a known graph structure and one selected real world dataset.
6. Performance analysis For the performance analysis of model will evaluate the model using suitable metrics such as prediction accuracy F1 score MAE or RMSE.

V. Dataset(s)

A. Lung Cancer Dataset

Source: Kaggle
URL: Kaggle Lung Cancer Dataset↗

The Lung Cancer dataset consists of 309 patient records with 16 features representing demographic information, symptoms, and lifestyle-related factors[2]. The dataset primarily contains binary variables, with gender and age represented separately. The binary variables use the following encoding:

- YES = 2
- NO = 1

The features included in the dataset are:

- GENDER
- AGE
- SMOKING
- YELLOW_FINGERS
- ANXIETY
- PEER_PRESSURE
- CHRONIC DISEASE
- FATIGUE
- ALLERGY
- WHEEZING
- ALCOHOL CONSUMING
- COUGHING
- SHORTNESS OF BREATH
- SWALLOWING DIFFICULTY
- CHEST PAIN
- LUNG_CANCER

The dataset is compact and contains a relatively large number of variables compared to the number of patient records, making it suitable for efficient processing, model validation, and analysis of dependencies among healthcare-related variables.

B. Birth Cohort Linked Birth–Infant Death Dataset

Source: Centers for Disease Control and Prevention (CDC)
URL: CDC Vital Statistics Online↗

The Birth Cohort Linked Birth–Infant Death dataset contains information related to pregnancy, birth, and infant mortality[3]. The base paper uses the 2015 dataset, which contains approximately 3.99 million live births in the United States. The dataset includes demographic, maternal, pregnancy, delivery, and infant-related variables, making it suitable for studying complex dependencies among mixed-type healthcare variables.

C. Synthetic Gaussian Graphical Model Dataset

Source: Programmatically generated

A synthetic dataset will be generated using a predefined Gaussian Graphical Model (GGM) with a known dependency structure. The generated samples will be used to validate the NGM's ability to learn and preserve the underlying graphical dependencies. Graph recovery will be evaluated using metrics such as AUC and AUPR, following the synthetic experiment in the base paper[1].

VI. GitHub Repositories

The GitHub Repo whihc is the only official repo for this paper. The repo will be maintained throughout the project.

https://github.com/Priyam-Official/CSE-516-Group-1↗

REFERENCES

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